

Riemann X_i function using J-Math

let $X_i(s)$ denote the Riemann X_i function
that is, we have

$$X_i(s) = X_i(1 - s)$$

Let $s = a + bj$ such that

$$\{a, b \in C\}$$

i.e. Let $a = x_1 + y_1 i$

Let $b = x_2 + y_2 i$

$s = a + bj$ can be written as

$$s = [x_1 + y_1 i] + [x_2 + y_2 i]j$$

$$\boxed{s = x_1 + y_1 i + x_2 j + y_2 i j}$$

$\hookrightarrow$ i.e a 4D complex J-number.

Let $a = x_1 + y_1 i = \frac{1}{2} + y_1 i$

let b be the shift vector. i.e. $b = x_2 + y_2 i$ represents the horizontal shift (x_2) and vertical shift ($y_2 i$).

$$X_i(s) = X_i(1 - s)$$

let $LHS = X_i(s)$ and $RHS = X_i(1 - s)$

substituting $s = a + bj$, we get

$$LHS = X_i(a + bj) \text{ and } RHS = X_i(1 - a - bj)$$

Using GUGF from Jmath, we get

$$LHS = X_i(a + bj) = X_i(a) + [X_i(a + b) - X_i(a)]j$$

$$RHS = X_i(1 - a - bj) = X_i(1 - a) + [X_i(1 - a - b) - X_i(1 - a)]j$$

Let us put these Equations equal to zero, so that, we have roots of these equation. i.e. we get

$$LHS = 0 + 0j = X_i(a) + [X_i(a + b) - X_i(a)]j$$

$$RHS = 0 + 0j = X_i(1 - a) + [X_i(1 - a - b) - X_i(1 - a)]j$$

By comparison of J-coordinates, we get

LHS $\Rightarrow$

$$X_i(a) = 0$$

$$X_i(a + b) = 0$$

RHS $\Rightarrow$

$$X_i(1 - a) = 0$$

$$X_i(1 - a - b) = 0$$

From LHS and RHS, we get the following observations: From LHS $a = \frac{1}{2} + y_1 i$ is a zero of the X_i function then $1 - a$ is also a zero of the X_i function. so is $\tilde{a}$ and $1 - \tilde{a}$ i.e

$$a = 1 - \tilde{a} = \frac{1}{2} + y_1 i$$

$$1 - a = \tilde{a} = \frac{1}{2} - y_1 i$$

From RHS

$a + b = (\frac{1}{2} + x_2) + (y_1 + y_2)i$ is a zero of the X_i function
then $1 - a - b = (\frac{1}{2} - x_2) - (y_1 + y_2)i$ is also a zero of the X_i function, i.e. we get

$$a + b = (\frac{1}{2} + x_2) + (y_1 + y_2)i$$

$$1 - a - b = (\frac{1}{2} - x_2) - (y_1 + y_2)i$$

$$\widetilde{a + b} = (\frac{1}{2} + x_2) - (y_1 + y_2)i$$

$$1 - \widetilde{a - b} = (\frac{1}{2} - x_2) + (y_1 + y_2)i$$

Important Note

In $s = a + bj$
OR

$$s = \frac{1}{2} + x_2j + y_1i + y_2ij$$

- ① a is the baseline anchor.
- ② b is the shift vector.
- ③ x_2 = Horizontal shift from the baseline anchor.
- ④ y_2i = Vertical shift from the baseline anchor.
- ⑤ x_2 is responsible for horizontal shift in $s = \frac{1}{2} + x_2j + y_1i + y_2ij$
- ⑥ y_2i is responsible for vertical shift in $s = \frac{1}{2} + x_2j + y_1i + y_2ij$
- ⑦ Since a is the baseline zero of the X_i function, then adding the shift b to it, represents the shifted zero ($a + b$)

Theorem to be proven

x_2 is zero

Proof

Let us assume, for the sake of the argument that x_2 is not zero. i.e $x_2 \neq 0$

① x_2 not being zero geometrically and algebraically means that The horizontal shift vector (x_2) is not zero in

$$s = \frac{1}{2} + x_2j + y_1i + y_2ij$$

② Let A, B, C act as formal unevaluated symbolic intermediaries operating within the syntactic domain of J-Math's formal symbolic space. Let it be noted that the specific axioms of J-Math are defined and applies to semantics, not syntax. since the specific axioms of J-Math do not apply to syntactics, we are free to choose syntactic axioms, of symbolic J-space. Therefore, we choose standard structural axioms of algebra, for this symbolic space. hence Associativity, commutativity and distributivity holds unconditionally. Under the symbolic space, the following identity is established.

$$\boxed{C \cdot [A + B] = C \cdot A + C \cdot B}$$

③ We now assign these formal symbolic intermediaries to our specific J-Math variable and evaluated Riemann roots. i.e. Let C map to x_2j

Let A map to $X_i(r_1)$

Let B map to $X_i(r_2)$

where r_1 and r_2 are roots of the Riemann X_i function. Therefore $X_i(r_1)$ and $X_i(r_2)$ will ultimately map / evaluate to " $0 + 0j$ ". These intermediaries will map to their assigned values via an evaluation map.

④ Let " $ev(x)$ " be the evaluation map that bridges/evaluates syntactic identity into the semantic space of J-Math. We now use the evaluation map to the identity established earlier.

From LHS $\Rightarrow ev[C \cdot (A + B)]$
here by evaluation, we get

$$x_2 J \cdot [X_i(r_1) + X_i(r_2)]$$

since $X_i(r_1)$ and $X_i(r_2)$ evaluates to $0 + 0j$, we get

$$x_2 J \cdot [0 + 0J]$$

By the axiom " $0 \cdot (a + bj) = 0$ " of J math, we get

$$\boxed{LHS = x_2}$$

From RHS $\Rightarrow ev[C \cdot A + C \cdot B]$
here by evaluation, we get

$$x_2 J \cdot (0 + 0j) + x_2 J \cdot (0 + 0j)$$

By axiom " $0 \cdot (a + bj) = 0$ " of J-Math, we get

$$x_2 + x_2$$

$$\boxed{RHS = 2x_2}$$

⑤ By Leibniz's Law [The Identity of indiscernibles], the LHS and RHS are structurally identical mathematical entities within the syntactic domain.

Furthermore, since evaluation map is a function. By the definition of function, it must give one unique output to identical input. i.e. since LHS and RHS are identical entities, within the syntactic domain evaluating these identical inputs must give one unique output. i.e. If $X = Y$ then $f(X) = f(Y)$

Therefore, because LHS = RHS syntactically

then $ev(LHS) = ev(RHS)$

i.e. we get from LHS and RHS

$$x_2 = 2x_2$$

solving for x_2 , gives the only valid real semantic solution, that is zero. i.e.

$$x_2 = 2x_2$$

$$-x_2 \quad -x_2$$

$$\boxed{0 = x_2}$$

Conclusion

The evaluation map structurally dictates that $x_2 = 0$. However our initial assumption was strictly that $x_2 \neq 0$

This creates a mathematical state where $x_2 = 0$ and $x_2 \neq 0$ exist simultaneously in the same respect. By the 'Law of Non-contradiction' a proposition and its negation cannot both be true simultaneously in the same respect. Hence our assumption that lead to this Paradox, must be invalid. i.e. Assumption that

$$x_2 \neq 0$$

is invalid

Therefore $\boxed{x_2 = 0}$

because $x_2 = 0$, this geometrically and algebraically means that, the horizontal shift vector, that was responsible for horizontal shift from the anchor, is zero. i.e. No horizontal shift from the baseline anchor is possible.

Q.E.D.